

# RETAIL BUSINESS PERFORMANCE & PROFITABILITY ANALYSIS

## 1. Introduction

In modern retail ecosystems, unoptimized inventory lifecycles and opaque supply chain disruptions present severe threats to operational profitability. This project was initiated to audit transactional sales data, isolate the financial impact of prolonged warehouse holding durations, and identify seasonal demand fluctuations.

## 2. Abstract

A comprehensive analytics pipeline was engineered utilizing Python, MySQL, and Tableau. 10,000 retail transactions were generated and piped through a relational database to sanitize missing records and compute exact net margins. Advanced statistical models—specifically Pearson correlation and K-Means clustering—were then applied to classify product inventory into three distinct risk tiers. The findings mathematically prove that extended holding durations trigger aggressive markdown cycles.

## 3. Tools Used

- **Python (Pandas, Numpy, Scikit-Learn, Matplotlib/Seaborn):** Deployed for synthetic data engineering, logistics simulation, correlation plotting, and predictive machine learning.
- **MySQL Workbench:** Utilized as the enterprise data warehouse to process cleansing commands, handle NULL values, and construct complex financial aggregation views.
- **Tableau Public:** Deployed to build an interactive Business Intelligence dashboard featuring interconnected, parameter-driven visual architectures.

#### 4. Steps Involved in Building the Project

- **Step 1: Data Generation:** Engineered an enterprise-scale dataset in Python, deliberately injecting supply chain variance and missing sub-categories to rigorously test downstream database hygiene.
- **Step 2: Database Ingestion & Cleansing:** Piped raw data into MySQL. Executed SQL queries to standardize unstructured text and dynamically calculate `net_profit` and `profit_margin_pct` via relational views.
- **Step 3: Machine Learning & Visualization:** Extracted the cleaned data back into Python. Calculated a negative Pearson correlation coefficient between `inventory_days` and `profit_margin`. Applied a K-Means algorithm to group products into three risk tiers based on mathematical holding behaviors.
- **Step 4: BI Architecture:** Connected the finalized dataset to Tableau, constructing an interactive visual architecture containing dense Column, Bar, and Histogram graphs linked via cross-filtering action parameters.

#### 5. Conclusion

The empirical data dictates an immediate shift in inventory strategy. The negative correlation identified by the Python algorithm proves that stagnant inventory triggers forced liquidation, eroding gross profits. The AI clustering model successfully identified that supply chain delays push otherwise profitable assets into critical risk tiers. Operations must adopt Just-In-Time (JIT) procurement for highly elastic items and utilize strategic product bundling to efficiently clear aging warehouse stock.